# mark-academic-slide

One template for XeLaTeX and reveal.js

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# Outline

Text and blocks

Figures and tables

Equations and code

# Text and blocks

# Lists and emphasis

- Plain text is Noto Sans, with vermillion for alerts and *italics* for terms
- 繁體中文可直接輸入，technical terms 保留 English
- Nested lists
  - Second level keeps the same line spacing
  - Third level is rarely needed
- Numbered steps
  1. Align reads to the reference
  2. Call variants and normalize them
  3. Annotate and filter

## Blocks

**Definition**

The Phred score $Q$ of a base call is $Q = -10 \log_{10} P_{\text{error}}$.

**Warning**

Check the genome build before you merge coordinates from two tables.

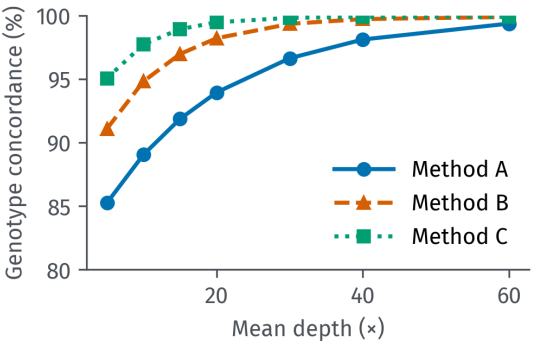
**Example**

$Q = 30$ corresponds to one expected error in 1,000 base calls.

# Figures and tables

# Figure with text

- Concordance rises with depth
- Method C reaches 99% at 20×
- Lines differ by color, marker, and dash




Simulated data, for layout only.

# Table

| Caller | Precision (%) | Recall (%) | F1 (%) |
|---|---|---|---|
| Method A | 97.1 | 92.4 | 94.7 |
| Method B | 98.3 | 95.0 | 96.6 |
| Method C | 99.0 | 97.8 | 98.4 |

Simulated values for layout only.

# Equations and code

# Equations

Hardy-Weinberg genotype frequencies for allele frequencies $p + q = 1$:

$$p^2 + 2pq + q^2 = 1$$

Posterior probability of genotype $G$ given reads $D$:

$$P(G \mid D) = \frac{P(D \mid G)\, P(G)}{\sum_g P(D \mid g)\, P(g)}$$

$$F_1 = \frac{2\,\mathsf{TP}}{2\,\mathsf{TP} + \mathsf{FP} + \mathsf{FN}}$$

## Code

```python
import pysam

def count_pass(path: str, min_qual: float = 30.0) -> int:
    # Count PASS records with QUAL >= min_qual
    with pysam.VariantFile(path) as vcf:
        return sum(1 for rec in vcf if rec.qual >= min_qual)
```

```bash
# Keep PASS variants and index the result
bcftools view -f PASS -Oz -o pass.vcf.gz calls.vcf.gz
tabix -p vcf pass.vcf.gz
```

# Palette



Orange
#E69F00



Sky blue
#56B4E9



Bluish green
#009E73



Yellow
#F0E442



Blue
#0072B2



Vermillion
#D55E00



Reddish purple
#CC79A7



Black
#000000

# Thank you